DO NOW — RECALL

Lesson 076 · Algorithm Choice: Time, Space and Precond

Name: _____ Class: _____ Date: _____

1. What must binary search assume?

2. Define auxiliary space.

3. Why can preprocessing change a choice?

4. Binary search in [1,3,5,7,9,11,13,15,17] for 15: give midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Data are sorted.

2. Model answer

Extra working memory beyond input storage.

3. Model answer

A one-time sort has cost.

4. Model answer

mid4=9, mid6=13, mid7=15; return index7.

By the end, you can…

• Choose an algorithm based on input properties, particularly whether data are already sorted.

• Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

• Sorting first takes time; consider that one-time cost and the number of future searches.

Vocabulary — know these terms

precondition — Input requirement for correctness.

preprocessing — Preparing data before main operation.

amortise — Spread one-off cost across tasks.

NEW LEARNING

Read, annotate and explain

Understand

Choose an algorithm based on input properties, particularly whether data are already sorted.

Trace

Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

Explain

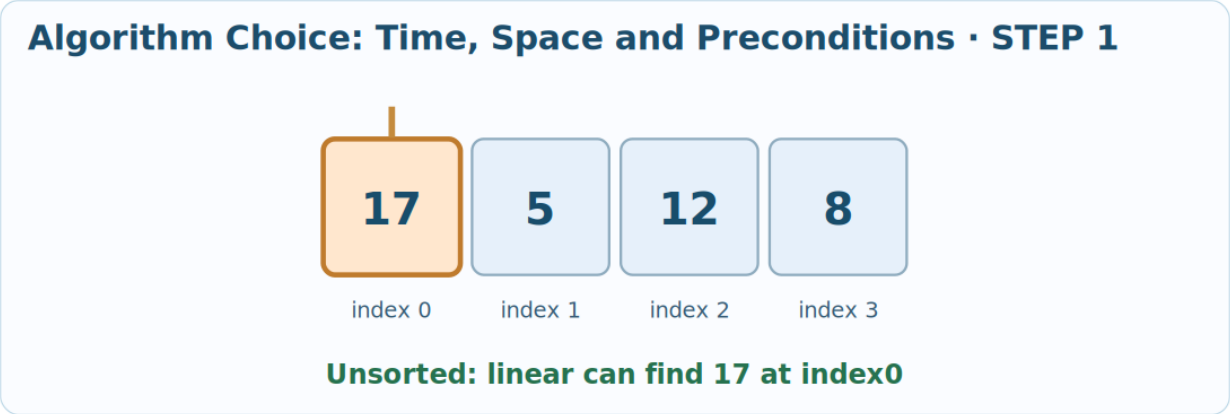
Sorting first takes time; consider that one-time cost and the number of future searches.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

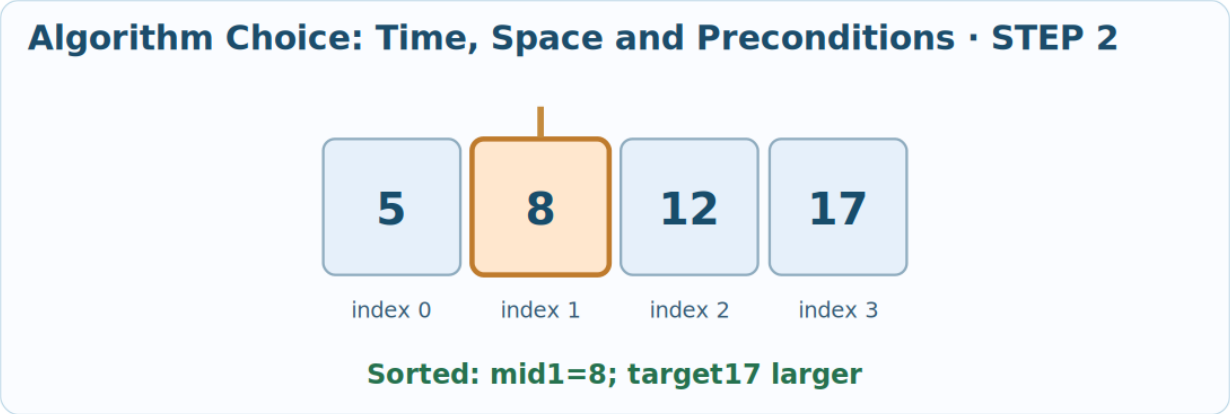
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Algorithm Choice: Time, Space and Preconditions · STEP 1; array 17, 5, 12, 8; Unsorted: linear can find 17 at index0

New Learning

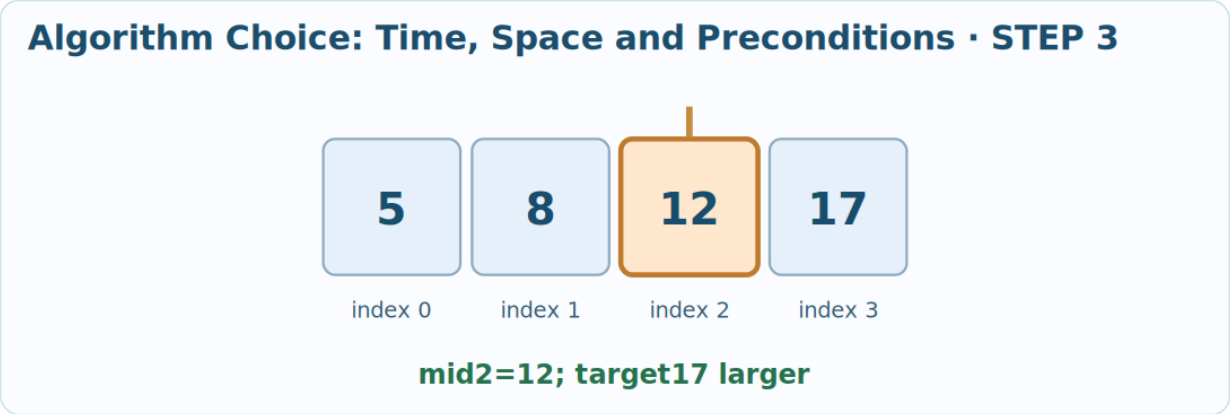


Algorithm Choice: Time, Space and Preconditions · STEP 2; array 5, 8, 12, 17; Sorted: mid1=8; target17 larger

New Learning · original lesson diagrams

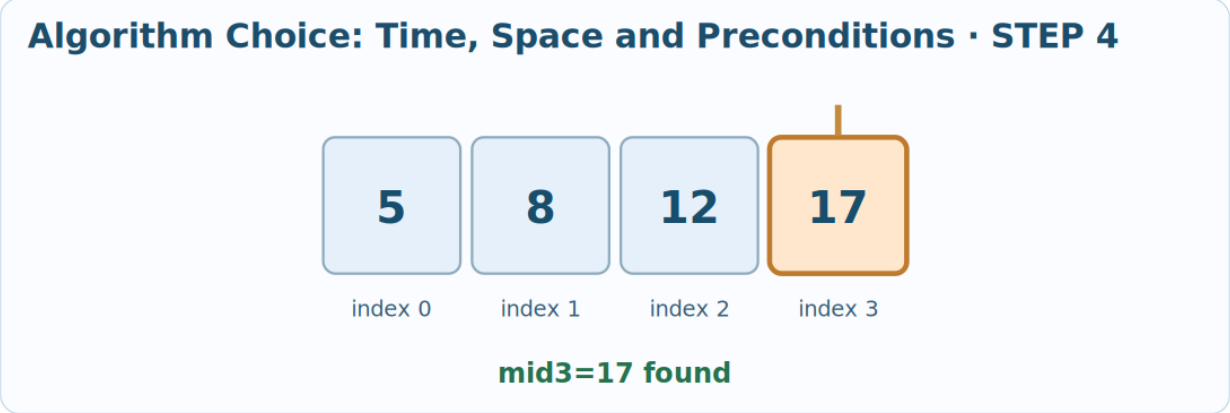
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Algorithm Choice: Time, Space and Preconditions · STEP 3; array 5, 8, 12, 17; mid2=12; target17 larger

New Learning



Algorithm Choice: Time, Space and Preconditions · STEP 4; array 5, 8, 12, 17; mid3=17 found

I DO — WORKED EXAMPLE

Follow the worked method

Choose a method to find 7 in unsorted [4,7,2].

Check the working

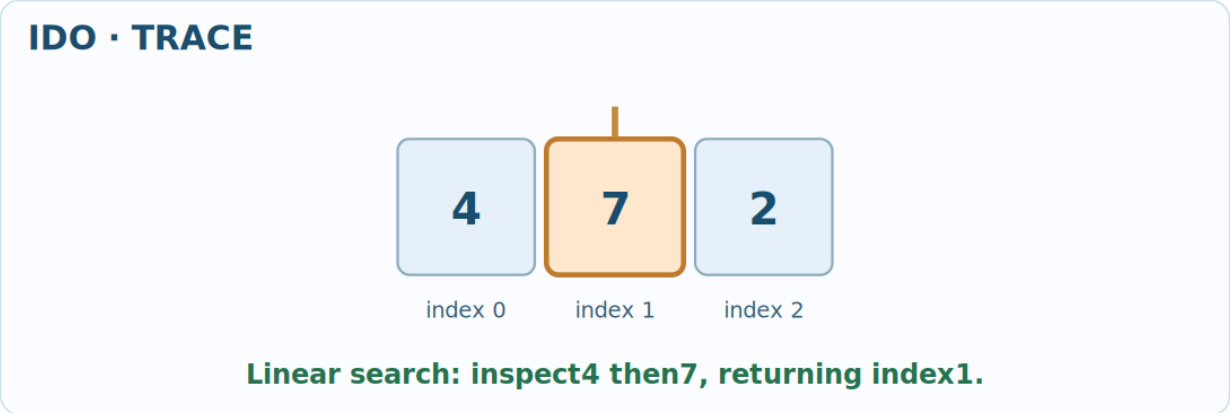
Linear search: inspect4 then7, returning index1.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose a method to find 7 in unsorted [4,7,2].

Linear search: inspect4 then7, returning index1.



IDO · TRACE; array 4, 7, 2; Linear search: inspect4 then7, returning index1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Find 12 in sorted [2,5,7,9,12,16,20] using binary search.
Source code / input:
mid3=9, mid5=16, mid4=12, return index4.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Choose linear/binary for unsorted [8,2,5,9], target5. [4]

Question 2

Trace binary search for 12 in [2,5,7,9,12,16,20]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare one search in short unsorted list versus many searches in large fixed dataset. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Binary search must always be preferred even for one lookup in unsorted data.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

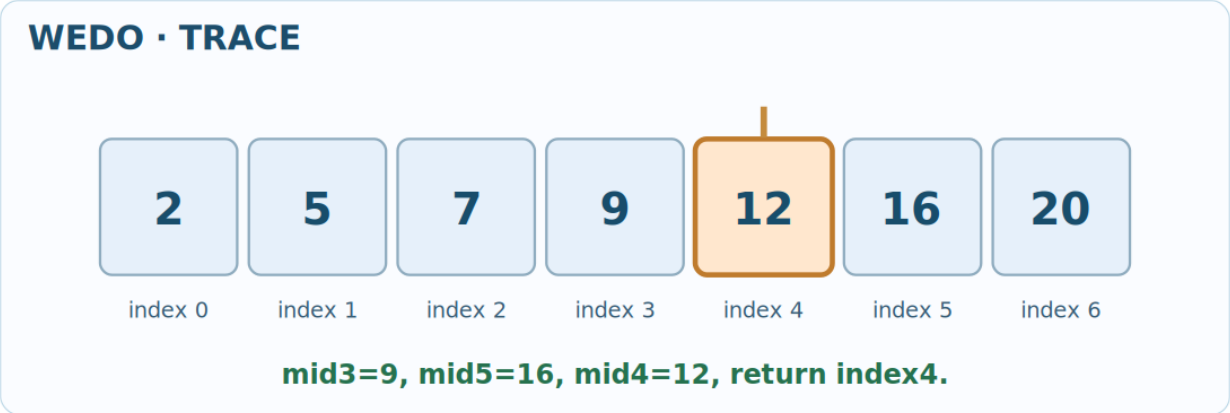
Check the working
mid3=9, mid5=16, mid4=12, return index4.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find 12 in sorted [2,5,7,9,12,16,20] using binary search.

mid3=9, mid5=16, mid4=12, return index4.



WEDO · TRACE; array 2, 5, 7, 9, 12, 16, 20; mid3=9, mid5=16, mid4=12, return index4.